Q1:NOV 2004 P2

8 **(a)** Three common emissions during radioactive decay are α-particles, β-particles and γ-rays.
Table 8.1 shows the properties of these emissions with respect to charge, approximate mass and effect of an electric field. Fill in the blank spaces.

Table 8.1

	charge	approximate mass	effect of an electric field
α-particle		4 units	deflected
β-particle	-1e		
γ-rays		0 units	not deflected

[4]

(b) The results of Rutherford's α-scattering experiment ushered a new model for the structure of an atom. Describe briefly this model.

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____ [3]

Q2:JUN 2018 P2

7 **(a)** Fig.7.1 shows a velocity selector.

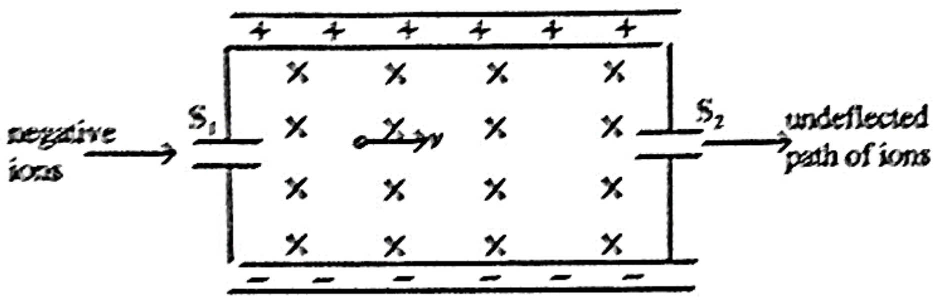

Fig.7.1

Negative ions enter the selector through S_1 and some leave through S_2.

(i) State the directions of the magnetic and electric forces on the ion.

magnetic ______________________________

____

electric ______________________________

____

(ii) Derive an expression of the speed v for ions that pass the selector undeflected.

(iii) The magnetic flux density B is 0.30 T and the electric field strength E is $1.5 \times 10^3 \text{Vm}^{-1}$.

Calculate the speed, v.

v = ______________________________ [5]

(b) Explain why ions travelling at a speed greater than the answer in **(a) (iii)** will not emerge from the selector.

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____

______________________________ [3]

Q3:JUN 2010 P5

4 (a) Explain what is meant by *quantisation of charge*. [1]

(b) State and explain what happens to the kinetic energy of a charged body moving through a uniform magnetic field. [2]

(c) In the Millikan experiment, to find the charge of an electron, e, the following results were obtained for various sizes of oil drops.

Experiment number	1	2	3	4	5	6
Charge/x10^{-19}C	11.35	12.96	3.24	4.82	6.40	6.42

Determine the value of e from these results. [4]

(d) (i) An electron emitted from a hot cathode in an evacuated tube is accelerated by a p.d. of 1.2×10^3 V.

Calculate the kinetic energy and the speed acquired by the electron.

(ii) Explain any changes that would occur if the electron is accelerated in air. [5]

Q4:NOV 2011 P5

4 (a) Millikan's oil drop experiment was a major success in the determination of the electron charge.

(i) State the evidence provided by the results of this experiment.

(ii) Briefly describe the principles involved in the determination of the electronic charge by Millikan's experiment.

(iii) To improve accuracy in the experiment, Millikan used a heat jacket and a less volatile oil.

Explain how each of these improved the accuracy of the results.

(iv) A student investigating the magnitude of the electronic charge, based on non-S.I units, obtained the following data of charge on oil drops:

26.08; 22.82; 9.77; 26.08;16.30; 22.82; 19.56; 9.78; 16.30; 26.08; 9.78; 16.30; 19.56; 9.79

Use these results to determine the charge on an electron according to the student.

[10]

(b) **Fig. 4.1** shows a simplified form of a mass spectrometer.

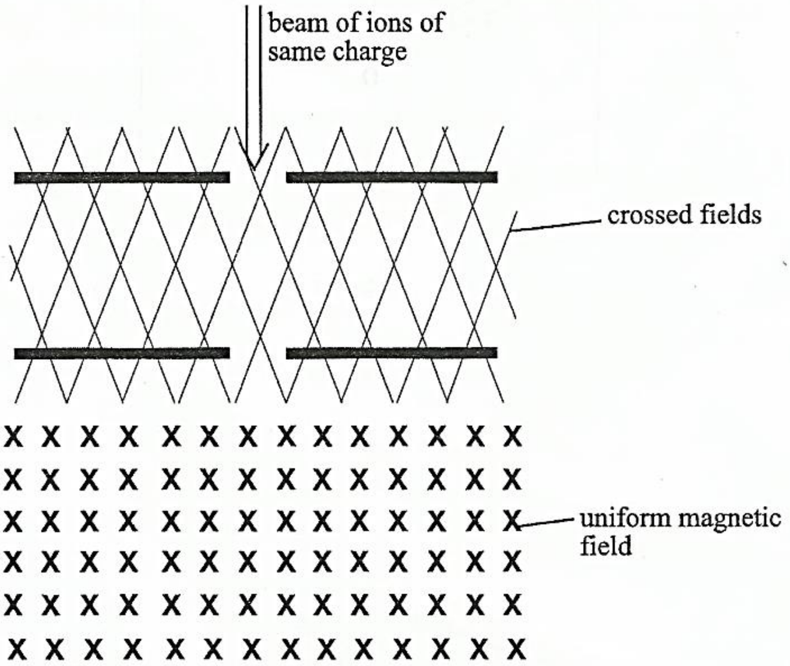

Fig. 4.1

Suggest

(i) the use of the crossed fields,

(ii) how the separation of ions of different masses is achieved.

[2]

Q5:JUN 2009 P5

3 **(a)** Oxygen atoms are ionised by the removal of electrons. The ions are accelerated in a vacuum and injected into a region where there is uniform magnetic and electric fields as shown in **Fig. 3.1**.

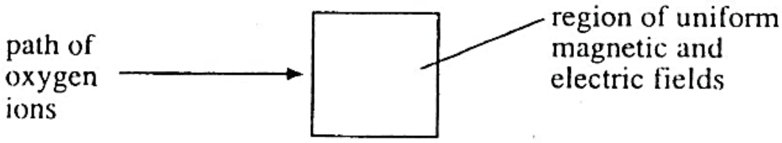

Fig. 3.1

(i) Describe the conditions necessary for the ions to pass through the fields undeflected.

(ii) If the speed of the ions as they reach the fields is 2.1×10^7 m/s and the magnetic flux density is 2.5×10^{-4} T, calculate the magnitude of the electric field strength so that the ions are not deflected in the region of the fields.

(iii) State the change, if any, in the path of the ions in the two fields if the ionisation of oxygen resulted in the removal of one electron.

[6]

Q6:JUN 2016 P5

2 (a) An oil drop, of mass m, is given a charge, Q, and is allowed to enter a uniform electric field. It is held stationary between two horizontal parallel plates by adjusting the electric field such that a potential difference, V, is between the plates. The charge, Q, is changed and the corresponding potential difference, V, to keep the drop stationary is obtained. **Fig. 2.1** shows the results.

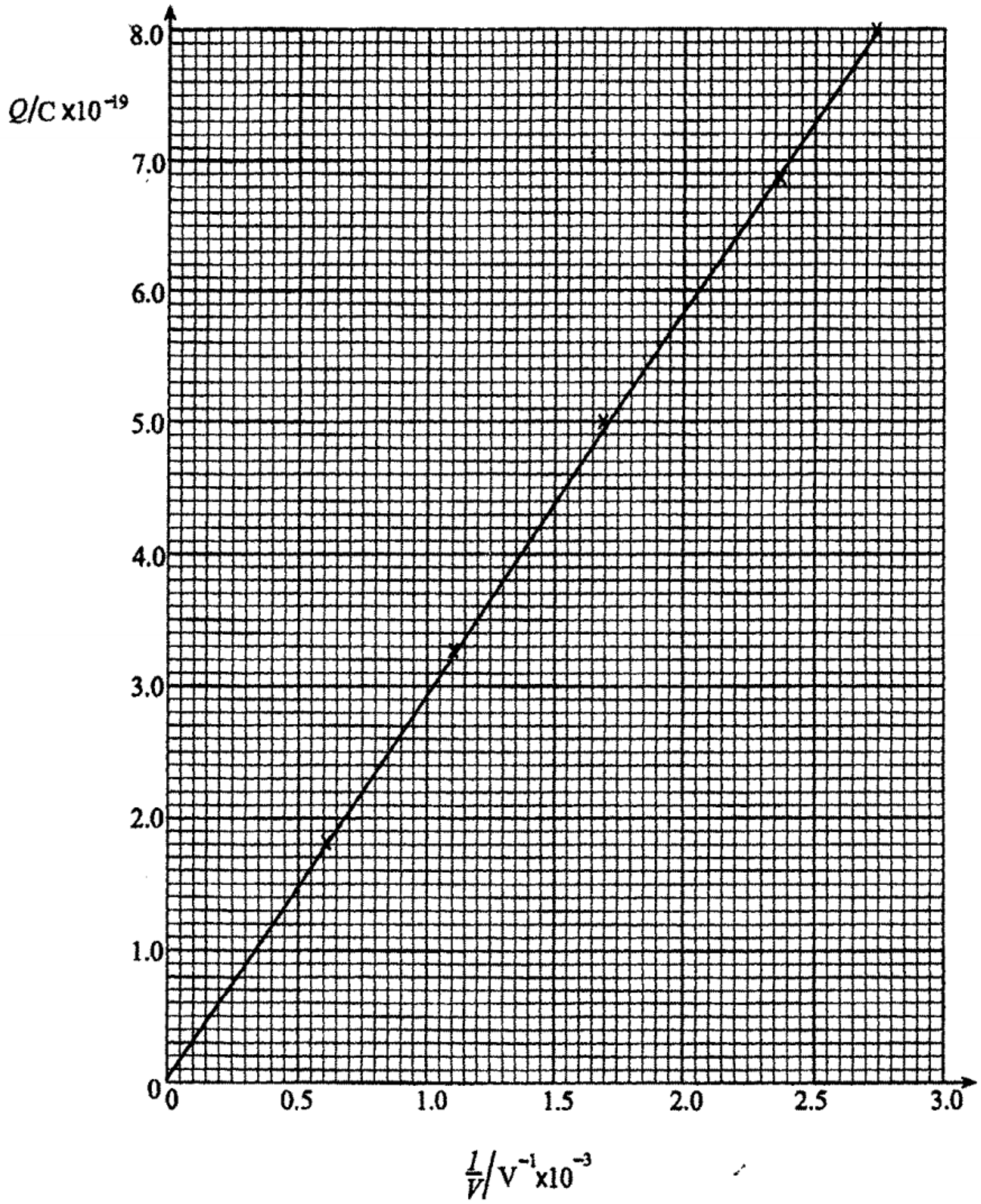

Fig. 2.1

(i) Show that Q is related to V and m by the expression

$$Q = \frac{0.196\, m}{V},$$

where the upthrust is negligible and the separation between the plates is 20.0 mm.

(ii) Use Fig. 2.1 to determine the mass of the oil drop. [7]

Q7:JUN 2017 P5

2 (a) (i) State two pieces of evidence from the results of Millikan's oil drop experiment supporting quantization of charge.

A negatively charged oil drop of mass 9.48×10^{-15} kg is held stationery in an electric field of strength $2.90 \times 10^{5}\ NC^{-1}$.

(ii) Determine the number of excess electrons on the oil drop. [4]

Q8:NOV 2014 P5

1 (a) (i) State the evidence provided by Millikan's oil drop experiment.

(ii) Explain how electric and magnetic fields are used in velocity selection. [4]

Q9:JUN 2013 P5

5 (a) Explain the term *Quantisation of charge*. [1]

(b) (i) Describe how the radius of an oil drop is determined in Millikan's oil drop experiment.

(ii) Fig. 5.1 shows a diagram of Millikan's oil drop experiment.

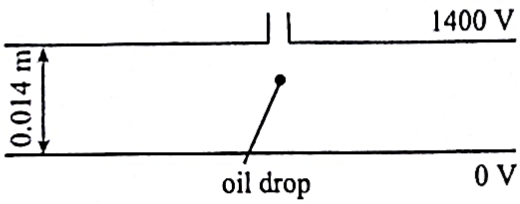

Fig. 5.1

Calculate the number of electrons which must be attached to the oil drop of mass 4.9×10^{-15} kg for it to remain stationary in the air between the plates (Assume the density of air is negligible compared to oil).

(iii) State with a reason the effect of the density of air being significant. [9]

(c) The following charges on various oil drops were found by the Millikan experiment:

26.4×10^{-19} C; 4.4×10^{-19} C; 13.2×10^{-19} C; 11.0×10^{-19} C

Deduce the value of the charge of an electron. [2]

Q10:NOV 2007 P3

7 A beam of electrons enters a uniform electric field horizontally at 1.5 cm from the lower plate as in **Fig. 7.1**. The separation of the plates is 7.8 cm. The upper plate is maintained at a potential of 3 400 V with respect to the lower plate. The length of each plate is 20.0 cm. The speed of a particular electron in the beam is 6.7×10^7 m/s.

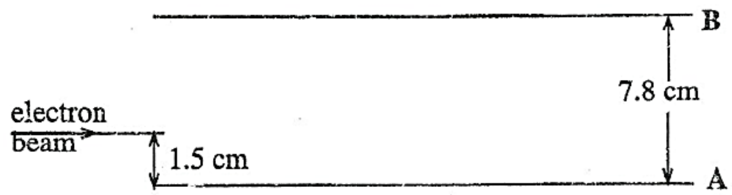

Fig. 7.1

(i) (a) Describe and explain the path taken by the beam in the electric field. [2]

(b) Calculate

(i) the electric field strength between the plates,

(ii) the time in which the electron remains in the field,

(iii) the vertical displacement at which the electron emerges from the field,

(iv) the velocity of the beam on emerging from the field,

(v) the direction of the electron on emerging from the field. [10]

(c) The electrons in the beam are travelling at different velocities.

(i) Describe the changes that should be done to **Fig. 7.1** which will enable electrons travelling at 6.7×10^7 m/s to be selected.

(ii) Suggest how the charge to mass ratio of an electron can be determined. [6]

(d) The beam of electrons is in a gravitational field.

Explain why force due to gravity is not considered in the calculations above. [2]